

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/11001088>

Correlates of adults' participation in physical activity: Review and update

Article in *Medicine & Science in Sports & Exercise* · January 2003

DOI: 10.1249/01.MSS.0000038974.76900.92 · Source: PubMed

CITATIONS

1,704

READS

37,378

5 authors, including:



Stewart Trost

Queensland University of Technology

324 PUBLICATIONS 21,972 CITATIONS

[SEE PROFILE](#)



Neville Owen

Baker Heart and Diabetes Institute

715 PUBLICATIONS 45,157 CITATIONS

[SEE PROFILE](#)



Wendy J Brown

The University of Queensland

615 PUBLICATIONS 17,869 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



CIHR Research: PI: Dr. Guy Faulkner [View project](#)



iSOPARC Project [View project](#)

Correlates of adults' participation in physical activity: review and update

STEWART G. TROST, NEVILLE OWEN, ADRIAN E. BAUMAN, JAMES F. SALLIS, and WENDY BROWN

School of Human Movement Studies; School of Population Health; University of Queensland, AUSTRALIA; School of Community Medicine, University of New South Wales, AUSTRALIA; and Department of Psychology, San Diego State University, San Diego, CA

ABSTRACT

TROST, S. G., N. OWEN, A. E. BAUMAN, J. F. SALLIS, and W. BROWN. Correlates of adults' participation in physical activity: review and update. *Med. Sci. Sports Exerc.*, Vol. 34, No. 12, pp. 1996–2001, 2002. **Purpose:** To review and update the evidence relating to the personal, social, and environmental factors associated with physical activity (PA) in adults. **Methods:** Systematic review of the peer-reviewed literature to identify papers published between 1998 and 2000 with PA (and including exercise and exercise adherence). Qualitative reports or case studies were not included. **Results:** Thirty-eight new studies were located. Most confirmed the existence of factors already known to be correlates of PA. Changes in status were noted in relation to the influence of marital status, obesity, smoking, lack of time, past exercise behavior, and eight environmental variables. New studies were located which focused on previously understudied population groups such as minorities, middle and older aged adults, and the disabled. **Conclusion:** The newly reported studies tend to take a broader "ecological" approach to understanding the correlates of PA and are more focused on environmental factors. There remains a need to better understand environmental influences and the factors that influence different types of PA. As most of the work in this field still relies on cross-sectional studies, longitudinal and intervention studies will be required if causal relationships are to be inferred. **Key Words:** DETERMINANTS, EXERCISE, LITERATURE REVIEW, DEMOGRAPHIC, PSYCHOLOGICAL, SOCIAL, ENVIRONMENTAL

I ncreasing participation in regular physical activity is a national health priority for many industrialized nations (13,44). Interventions are most effective when they alter the underlying variables that influence physical activity. Thus, studying "determinants" or correlates of physical activity is an important prerequisite for designing relevant policies and effective programs. Studies to date show the physical activity habits of adults to be associated with factors from multiple domains (37).

The goal of this paper is to review and update the research base on factors associated with physical activity in adults. For the purpose of highlighting recent progress and comparing current knowledge to the conclusions of earlier reviews, we also provide a summary in tabular form of those previously published reviews (17–19,36). We highlight the findings of studies published since the most recent comprehensive review, using methods and reporting procedures consistent with that review. The progress made is considered and priorities for future research are identified. Given the strong emphasis that national physical activity policy documents now place on the provision of opportunities for

being physically active, we pay particular attention to the new evidence on environmental factors related to physical activity.

EARLIER REVIEWS OF THE FINDINGS OF PHYSICAL ACTIVITY DETERMINANTS STUDIES

The most recent review of the adult physical activity "determinants" literature by Sallis and Owen (36) included 45 new studies published between 1992 and 1997. That review updated previous comprehensive reviews (17–19). Consequently, the findings from the approximately 300 studies on the determinants of adult physical activity were summarized. The authors concluded generally that adults' participation in physical activity is influenced by a diverse range of personal, social, and environmental factors. Of the six classes of "determinants" examined, individual-level variables such as socioeconomic status and perceived self-efficacy demonstrated the strongest and most consistent associations with physical activity behavior. In contrast, relatively few consistent positive or negative associations were found with respect to variables classified as behavioral attributes and skills, sociocultural influences, or physical environmental influences. However, fewer variables had been examined in these latter categories. The findings of the Sallis and Owen review are summarized in Table 1.

REVIEW AND UPDATE OF STUDIES PUBLISHED SINCE 1998

A search of the literature was conducted using several computer-based databases, including MEDLINE, PSYCLIT, SO-

Address for correspondence: Stewart G. Trost, Ph.D., School of Human Movement Studies, University of Queensland, Brisbane QLD 4072, Australia; E-mail: strost@hms.uq.edu.au.
Submitted for publication May 2001.
Accepted for publication July 2002.

TABLE 1. Updated summary of the factors associated with overall physical activity in adults.

Determinant	Sallis and Owen (36)	Updated Review	References	Change?
Demographic and biological factors				
Age	--	--	1,2,6,7,12,23,25,26,27,34,35,38,39,40,41,45	
Blue-collar occupation	-	-	39	
Childless	+	+	41	
Education	++	++	6,12,23,26,27,34,35,38,39,41,45	
Gender (male)	++	++	1,2,7,12,25,34,35,38,39	
Hereditary	++	++		
High risk for heart disease	-	-	12,35	
Income/socioeconomic status	++	++	1,3,4,6,7,12,35,38,39	
Injury history	+	+	40	
Marital status (married)	0	-	2,5,6,23,24,39,41	✓
Overweight/obesity	00	--	6,28,35,38,40,41	✓
Race/ethnicity (nonwhite)	--	--	6,12,23,40,41,45	
Psychological, cognitive, and emotional factors				
Attitudes	0	00	2,3,4,14,16,22,29,46	✓
Barriers to exercise	--	--	2,3,11,23,26,30,41,45	
Control over exercise	+	+	3,4,16,22,29,40	
Enjoyment of exercise	++	++	2,25	
Expect benefits	++	++	12,30,33,40	
Health locus of control	0	0		
Intention to exercise	++	++	3,4,14,16,22,29	
Knowledge of health and exercise	00	00	12,26,35	
Lack of time	-	--	2,23,41	✓
Mood disturbance	--	--		
Normative beliefs	00	00	3,4,14,16,22,24	
Perceived health or fitness	++	++	2,7,11,12,23,41	
Personality variables	+	+	14,15	
Poor body image	-	-		
Psychological health	+	+		
Self-efficacy	++	++	2,7,9,11,12,30,31,33,40,41,42,46	
Self-motivation	++	++	4,41	
Self-schemata for exercise	++	++	46	
Stage of change	++	++	33,42	
Stress	0	0	7,33	
Susceptibility to illness/seriousness of illness	00	00	7,35	
Value of exercise outcomes	0	0	7,8,33	
Behavioral attributes and skills				
Activity history during childhood/youth	00	0	43	✓
Activity history during adulthood	++	++	3,4,11,31	
Alcohol	0	0	21	
Contemporary exercise program	0	0		
Dietary habits (quality)	++	++	6,21,26	
Past exercise program	+	++	3,4,11,22,31	✓
Processes of change	++	++	42	
School sports	00	0	43	✓
Skills for coping with barriers	+	+		
Smoking	00	-	6,21,26,34,35,39,41	✓
Sports media use	0	0		
Type A behavior pattern	+	+		
Decisional balance sheet	+	+	42	
Social and cultural factors				
Class size				
Exercise models	0	0		
Group cohesion				
Past family influences	0	0		
Physician influence	++	++	7,12	
Social isolation	-	-	12,40	
Social support from friends/peers	++	++	2,8,9,16,20,25,41,45	
Social support from spouse/family	++	++	2,9,16,20,26,41,45	
Social support from staff/instructor				
Physical environment factors				
Access to facilities: actual	+	+		
Access to facilities: perceived	00	+	2,27	✓
Adequate lighting*		0	23,45	✓
Climate/season	--	--	1,23	
Cost of programs	0	0	2,27	
Disruptions in routine				
Enjoyable scenery*		+	1,23,45	✓
Frequently observe others exercising*		+	2,23,45	✓
Heavy traffic*		0	23,45	✓
Home equipment	0	+	2,23	✓
High Crime rates in the region*		0	10,23,45	✓
Hilly terrain*		+	23,44	✓
Neighborhood safety*		+	10,23,34,45	✓
Presence of sidewalks*		0	2,23,45	✓
Satisfaction with facilities*		+	2,27	✓
Unattended dogs*		0	23,44	✓
Urban location*		-	1,5,6,23,34,45	✓
Physical activity characteristics				
Intensity	-	-		
Perceived effort	--	--		

*, variable not examined in previous reviews; ++, repeatedly documented positive association with physical activity; +, weak or mixed evidence of positive association with physical activity; 00, repeatedly documented lack of association with physical activity; 0, weak or mixed evidence of no association with physical activity; --, repeatedly documented negative association with physical activity; -, weak or mixed evidence of negative association with physical activity.

Blank spaces indicate no data available.

cial science index, and sports discus. Manual searches were also made using the reference lists from recovered articles. Key words used for the computer searches were physical

activity, physical inactivity, exercise, determinants, correlates of physical activity, survey, health education, health behavior, and health promotion. Studies were included if the

dependent variable was physical activity, exercise, or exercise adherence, and if the study included participants aged 18 yr or older. Studies in which the dependent variable was aerobic fitness, intention, self-efficacy, or other intermediate (nonbehavioral) measures were not included. Qualitative reports or case studies were not included. Because this paper is an update on the most recent review, 1998 was chosen as starting point. Thirty-eight new studies published from 1998 to September 2000 were included in the review. Sample size for these studies ranged from 56 to 16,178, with a median of 1088. Notably, only 7 of the 38 studies (20.5%) utilized longitudinal study designs. None used objective measures of physical activity—all relied on self-report or attendance records. The self-report measures used in the new studies were comparable to those included in the 1998 review. Instruments ranged from single item global assessments to detailed activity inventories. The majority of studies focused on leisure time activity. In studies that reported multiple correlation coefficients, the proportion of variance accounted for ranged from 3% to 49.5%, with an average of $21.2\% \pm 15\%$.

To be consistent with the approach taken by previous reviews (17–19,36), factors associated with physical activity were classified as either a) demographic and biological; b) psychological, cognitive, and emotional; c) behavioral attributes and skills; d) social and cultural; e) physical environmental; or f) physical activity characteristics. Because only three new studies focused on adherence to structured exercise programs, the findings were summarized under the single heading of “overall physical activity.” Notably, the factors associated with adherence to structured exercise programs were also significant correlates of overall physical activity. Findings are summarized in Table 1.

DEMOGRAPHIC AND BIOLOGICAL FACTORS

Age and gender continued to be the two most consistent demographic correlates of physical activity behavior in adults. In studies that included men and women and that had sufficient age diversity to examine age-related trends, physical activity participation was consistently higher in men than in women and was inversely associated with age (1,2,6,7,12,23,25–27,31,34,35,38–41,45). Socioeconomic status, occupational status, and educational attainment were also consistent determinants of physical activity behavior (1,3,4,6,12,23,26,27,34,35,38,39,41,45). However, one study conducted in Australia found that inclusion of occupational and home activity eliminated the positive association between physical activity and occupational status in men. Among women, however, the inclusion of occupational and home activity had little effect on the association between occupational status and physical activity (39).

Studies examining the association between marital status and physical activity behavior produced mixed findings. Some studies reported a positive association between marital status and physical activity participation (5,24,39), others reported none (2,6,23,41). King et al. (24) examined the effects of marital transitions on changes in physical activity

in a cohort of men and women from the Stanford Five-City Project. The transition from a single to a married state resulted in significant positive changes in physical activity relative to individuals remaining single. In contrast, the transition from a married to a single state did not influence physical activity.

Overweight or obesity also emerged as a consistent negative influence on physical activity. Martinez-Gonzalez and colleagues (28) estimated the association between leisure time activity and weight status ($\text{BMI} > 30 \text{ kg} \cdot \text{m}^{-2}$) in a representative sample of the 15 member states of the European union. After controlling for hours spent sitting down, age, sex, education, recent weight change, social class, marital status, country of origin, and smoking, individuals in the highest quintile for leisure time physical activity were approximately 50% less likely than those in the lowest quintile to be classified as obese. Similar findings were reported in several other studies (6,35,38,40,41).

PSYCHOLOGICAL, COGNITIVE, AND EMOTIONAL FACTORS

Twenty-four new studies examined intrapersonal variables including attitudes, barriers to physical activity, enjoyment of physical activity, expected benefits, value of physical activity outcomes, intentions, exercise self-schemata (cognitive generalizations about the self in the context of exercise or physical activity), perceived behavioral control, normative beliefs, knowledge of health and exercise, perceived health, psychological health, self-efficacy, self-motivation, and stage of change (2–4, 6–9,11,12,14–16,22,23,26,29–31,33,35,41,42,45,46). Physical activity self-efficacy (a person's confidence in his or her ability to be physically active on a regular basis) emerged as the most consistent correlate of physical activity behavior (2,7,9,11,12,30,31,33,40–42,46). Oman and King (31) examined the influence of self-efficacy perceptions in a cohort of healthy sedentary men and women between the ages of 50 and 64. Among those participating in a supervised home-based activity program, baseline self-efficacy perceptions significantly predicted exercise adherence after 2 yr of follow-up. In a population-based study of 449 Australians aged 60 yr and older, Booth and colleagues (2) found self-efficacy to be strongly related to physical activity participation.

Sternfeld et al. (41) investigated the correlates of physical activity participation in an ethnically diverse sample of 2636 women enrolled in a North American health maintenance organization. Women with high levels of physical activity self-efficacy were two and four times more likely than were those with low levels of self-efficacy to be in the highest quartile for physical activity. However, Castro and colleagues (9) reported changes in self-efficacy and enjoyment to be inversely associated with activity change in physical activity behavior in 128 ethnic minority women.

Barriers to physical activity emerged as a strong influence (2,3,11,23,26,30,41,45). Lian et al. (26) assessed correlates of leisure-time activity in a population-based sample of

elderly men and women. Barriers to physical activity—lack of time, too tiring, too weak, fear of falling, bad weather, no facilities, and lack of exercise partners emerged as the strongest influence on leisure time activity for both men and women. In the U.S. Women's Determinants Study (6,20,23,45), perceived barriers of fatigue, ill health, lack of energy, and self-consciousness about appearance emerged as significant correlates of physical activity.

Constructs from the Theory of Reasoned Action and Theory of Planned Behavior (attitudes, normative beliefs, perceived behavioral control, and intentions) received relatively weak support (3,4,14,16,22,29). None of the studies found attitudes or normative beliefs to be associated with physical activity behavior. Consistent with previous reviews, knowledge related to health and physical activity was not associated with physical activity (12,26,35).

BEHAVIORAL ATTRIBUTES AND SKILLS

Dietary habits, past exercise behavior, smoking status, and decisional balance were the only behavioral attributes and skills examined in the new studies. Past exercise behavior or exercise habit emerged as a consistent predictor of current activity status (3,4,11,22,31). There were positive associations with healthy diet; however, only a small number of studies examined dietary behavior (6,21,26). With one exception (41), being a smoker was inversely related to physical activity (6,21,26,34,35,39).

SOCIAL AND CULTURAL FACTORS

Social support emerged as a consistently important correlate. Every study that included a measure of social support for physical activity found a significant positive association (2,8,9,16,20,25,26,41,45). Leslie et al. (25) studied the association with social support in a large sample of Australian college students. Those reporting low levels of social support from either family or friends were 23–55% more likely to be insufficiently active for health benefits than were those with reporting high levels of support. In the U.S. Women's Determinants Study, social support was strongly associated with physical activity. Women with high levels of physical activity social support were approximately twice as likely as were women with low support to be active at least 30 min on 5 or more days of the week (20).

PHYSICAL ENVIRONMENT FACTORS

In contrast to the studies published before 1998, recently published studies have included measures of physical environmental factors. Although the strength and direction of the associations with physical activity varied from study to study, there was sufficient evidence to identify several new environmental correlates of physical activity. These were individual level influences such as exercise equipment at home, access to facilities, satisfaction with recreation facilities, and community level influences such as neighborhood safety, hilly terrain, frequent observation of others engaging

in physical activity (modeling), and enjoyable scenery (2,10,23,27,34,45).

In their study of Australian adults aged 60 yr and over, Booth and colleagues (2) found that having friends who participated regularly in physical activity, safe footpaths for walking, and having access to a park were significantly associated with regular physical activity. MacDougall et al. (27) examined the influence of satisfaction with recreational facilities on physical activity behavior. After controlling for age, education, and health status, dissatisfaction with local recreational facilities was significantly associated with a greater risk of inactivity.

King et al. (23) examined the hypothesized environmental determinants of physical activity in a national sample of U.S. women aged 40 yr and older. Enjoyable scenery while exercising and frequently observing others exercise were positively associated with physical activity participation. Somewhat surprisingly, the presence of hills and unattended dogs were associated with more rather than less activity. This finding highlights the difficulty associated with making unambiguous inferences about perceived environmental influences from cross-sectional studies. For example, those who are active outdoors may be more aware of environmental barriers. In this study (23), other neighborhood characteristics such as perceived safety, presence of sidewalks, heavy traffic, adequate lighting, and high crime were not associated with physical activity status. In contrast, the Centers for Disease Control and Prevention (10) reported a significant positive association between perceived neighborhood safety and physical activity.

Six new studies examined the impact of urban location on physical activity participation (1,5,6,23,34,45). Notably, all of them found physical activity to be significantly lower among adults living in rural areas than in urban study participants, although most of the studies assessed leisure-time physical activity and not occupational physical activity.

DISCUSSION

Overall, the 38 new studies published between 1998 and September 2000 contribute significantly to the understanding of the factors associated with adults' physical activity. Somewhat surprisingly, most of the studies published during this period continued to examine variables for which there was very well established evidence of a negative or positive association with physical activity behavior. However, nearly all the new studies investigated previously understudied populations such as minorities, middle- and older-aged adults, and the disabled. The fact that similar results have now been observed in these population groups is an important development and has important implications for the work of policy makers and public health professionals.

Despite the recommendations of earlier reviews (17,18,19,36), only 7 of the 38 new studies used prospective study designs (4,7–9,24,31,42). In general, the results of these studies were consistent with the findings of cross-sectional studies. There were, however, two notable exceptions. In the 10-yr follow-up of participants in the Stanford Five City Project, King et al. (24) observed getting married

to be positively associated with physical activity over time. This finding was in contrast to the majority of cross-sectional studies in which marriage was inversely associated with physical activity. Another study that opposed the findings of cross-sectional studies was the study by Castro and colleagues (9). In that study, changes in exercise self-efficacy were inversely and not positively associated with changes in walking for exercise.

There has been progress in the area of environmental factors associated with activity, with the addition of 10 new variables to this section of the summary table. This trend reflects the increased use of ecological and broader health promotion models to explain physical activity behavior (32,36). The influence of the physical environment upon participation remains a high priority area for future research. The available evidence, although limited, suggested that access to facilities, satisfaction with facilities, neighborhood safety, access to exercise equipment at home, and frequently observing others exercise may be important factors.

Other notable changes from previous reviews were the findings in relation to marital status, overweight and obesity, attitudes, lack of time, past exercise behavior, and smoking status. Based on the most recent evidence, being married emerged as a weak yet negative influence on physical activity participation, whereas weight status and smoking changed from being repeatedly demonstrated nonassociations to repeatedly documented negative associations with physical activity. There was sufficient evidence to “upgrade” the classification of lack of time and past exercise behavior from weak or mixed evidence of association to a repeatedly documented association. Activity history during childhood and youth and school sports was upgraded from repeatedly documented lack of association to weak or mixed evidence of no association with adult activity. There was consistent lack of support for attitudes toward physical activity. This resulted in a change in classification for this variable from mixed or weak evidence of no association to repeatedly documented lack of association.

The blank spaces and equivocal associations shown in Table 1 highlight the need for more research in a number of domains. In particular, there is clearly a need to gain a better understanding of the specific physical environmental attributes that might influence physical activity, and how these attributes interact with known psychosocial influences of activity behavior. However, for progress to be made in this important area, more work is needed with respect to the measurement of environmental variables (37). There are unique challenges in studying environmental correlates of physical activity. Perhaps the most basic challenge is that many environmental variables are less subject to experimen-

tal control and manipulation than are, for example, intrapersonal variables. The fact that environmental variables are ubiquitous and can have widespread effects on the population makes them very difficult to study. If, for example, virtually every adult in a population owns at least one television and one motor vehicle, it is a challenge to document the effects of these variables.

Although new conceptual explanations of environmental influences on physical activity are a priority, the primary need is for empirical data. There is a need to determine whether environmental measures add variance to the explanation of behavior, above that provided by intrapersonal and social and cultural domains (37). Multiple geographic and cultural settings may need to be studied to achieve sufficient variation in environmental characteristics to study their associations with behavior. For some variables, it may be necessary to conduct studies in multiple nations. Ultimately, there will be a need to take the variables identified in correlational studies and target them for change in intervention studies. Ideally, multi-level interventions will be evaluated, and in some cases it may be possible to determine whether multilevel approaches that include environmental changes improve outcomes over what can be achieved by programs targeting only intrapersonal and interpersonal mediators (37).

As few of the studies examined the determinants of physical activity in different contexts, there is a need for future studies to identify the determinants of physical activities and sedentary behaviors in the context of work and daily living (32). That is, we need to know whether the determinants are different for transport activity (walking or cycling to and from places, leisure time activity, occupational activity, and incidental activity). We need more information about physical activity patterns at different life stages. In particular, we need to learn more about the role of pregnancy, childbirth, and parenting as a barrier to activity participation. We also need to learn more about the correlates of moderate-intensity physical activity and about the interactions between different influences. Such information is crucial if public education programs emphasizing lifestyle physical activity are to be pursued with success. Finally, the physical activity determinants literature is still predominantly based on cross-sectional studies, precluding the ability to infer causal relationships between the hypothesized determinants and physical activity. Accordingly, longitudinal and intervention studies in this area are needed.

Funding for this project was provided by the Commonwealth Department of Health and Aged Care, Canberra, Australia.

REFERENCES

1. BAUMAN, A., B. SMITH, L. STOKER L., B. BELLEW, and M. BOOTH. Geographical influences upon physical activity participation: evidence of a “coastal effect.” *Aust. N. Z. J. Public Health* 23:322–324, 1991.
2. BOOTH, M., N. OWEN, A. BAUMAN, O. CLAVISI, and E. LESLIE. Social-cognitive and perceived environment influences associated with physical activity in older Australians. *Prev. Med.* 31:15–22, 2000.
3. BOZIOELOS, G., and P. BENNETT. The theory of planned behavior as predictor of exercise: the moderating influence of beliefs and personality variables. *J. Health Psychol.* 4:517–529, 1999.
4. BRENES, G. A., M. J. STRUBE, and M. STORANDT. An application of the theory of planned behavior to exercise among older adults. *J. Appl. Soc. Psychol.* 28:2274–2290, 1998.

5. BROWN, W. J., A. F. YOUNG, and J. E. BYLES. Tyranny of distance? The health of mid-age women living in five geographical areas of Australia. *Aust. J. Rural Health* 7:148–154, 1999.
6. BROWNSON, R. C., A. A. EYLER, A. C. KING, D. R. BROWN, Y. L. SHYU, and J. F. SALLIS. Patterns and correlates of physical activity among US women 40 years and older. *Am. J. Public Health* 90:264–270, 2000.
7. BURTON, L. C., S. SHAPIRO, and P. S. GERMAN. Determinants of physical activity initiation and maintenance among community-dwelling older persons. *Prev. Med.* 29:422–430, 1999.
8. CASERTA, M. S., and P. A. GILLET. Older women's feelings about exercise and their adherence to an aerobic regimen over time. *The Gerontologist* 38:602–609, 1998.
9. CASTRO, C. M., J. F. SALLIS, S. A. HICKMAN, R. E. LEE, and A. H. CHEN. A prospective study of psychosocial correlates of physical activity for ethnic minority women. *Psychol. Health* 14:277–293, 1999.
10. CENTERS FOR DISEASE CONTROL AND PREVENTION. Neighborhood safety and the prevalence of physical inactivity—selected states, 1996. *MMWR* 48:143–146, 1999.
11. CLARK, D. O., and F. NORTHWEHR. Exercise self-efficacy and its correlates among socioeconomically disadvantaged older adults. *Health Educ. Behav.* 26:535–546, 1999.
12. CLARK, D. O. Physical activity and its correlates among urban primary care patients aged 55 years or older. *J. Gerontol.* 54B: S41–S48, 1999.
13. COMMONWEALTH DEPARTMENT OF HEALTH AND FAMILY SERVICES (DHFS). Developing an Active Australia: a framework for action for physical activity and health. DHFS: Canberra, 1998, pp. 1–18.
14. COURNEYA, K. S., T. M. BOBICK, and R. J. SCHINKE. Does the theory of planned behavior mediate the relation between personality and exercise behavior. *Basic Appl. Soc. Psychol.* 21:317–324, 1999.
15. COURNEYA, K. S., and L. M. HELLSTEN. Personality correlates of exercise behavior, motives, barriers and preferences: an application of the five-factor model. *Pers. Individ. Diff.* 24:625–633, 1998.
16. COURNEYA, K. S., R. C. PLOTNIKOFF, S. B. HOTZ, and N. J. BIRKETT. Social support and the theory of planned behavior in the exercise domain. *Am. J. Health Behav.* 24:300–308, 2000.
17. DISHMAN, R. K. Determinants of participation in physical activity. In: *Exercise, Fitness, and Health: A Consensus of Current Knowledge*, C. Bouchard, R. J. Shephard, T. Stephens, J. R. Sutton, and B. D. McPherson (Eds.). Champaign, IL: Human Kinetics, 1990, pp. 75–102.
18. DISHMAN, R. K., J. F. SALLIS, and D. R. ORENSTEIN. The determinants of physical activity and exercise. *Public Health Rep.* 100: 158–171, 1985.
19. DISHMAN, R. K., and J. F. SALLIS. Determinants and interventions for physical activity and exercise. In: *Physical activity, fitness and health: International proceedings and consensus statement*, C. Bouchard, R. J. Shephard, and T. Stephens (Eds.). Champaign, IL: Human Kinetics, 1994, pp. 214–238.
20. EYLER, A. A., R. C. BROWNSON, R. J. DONATELLE, A. C. KING, D. BROWN, and J. F. SALLIS. Physical activity social support and middle- and older-aged minority women: results from a US survey. *Soc. Sci. Med.* 49:871–789, 1999.
21. JOHNSON, M. F., J. F. NICHOLS, J. F. SALLIS, K. J. CALFAS, and M. F. HOVELL. Interrelationships between physical activity and other health behaviors among university women and men. *Prev. Med.* 27:536–544, 1998.
22. KERNER, M. S., and A. H. GROSSMAN. Attitudinal, social, and practical correlates to fitness behavior: a test of the theory of planned behavior. *Percept. Mot. Skills* 87:1139–1154, 1998.
23. KING, A. C., C. CASTRO, S. WILCOX, A. A. EYLER, J. F. SALLIS, and R. C. BROWNSON. Personal and environmental factors associated with physical inactivity among different racial-ethnic groups of US middle-aged and older aged adults. *Health Psychol.* 19:354–364, 2000.
24. KING, A. C., M. KIERNAN, D. K. AHN, and S. WILCOX. The effects of marital transitions on changes in physical activity: results from a 10-year community study. *Ann. Behav. Med.* 20:64–69, 1998.
25. LESLIE, E., N. OWEN, J. SALMON, A. BAUMAN, J. F. SALLIS, and S. K. LO. Insufficiently active Australian college students: perceived personal, social, and environmental influences. *Prev. Med.* 28:20–27, 1999.
26. LIAN, W. M., G. L. GAN, C. H. PIN, S. WEE, and H. C. YE. Correlates of leisure-time physical activity in an elderly population in Singapore. *Am. J. Public Health* 89:1578–1580, 1999.
27. MACDOUGALL, C., R. COOKE, N. OWEN, K. WILLSON, and A. BAUMAN. Relating physical activity to health status, social connections and community facilities. *Aust. N. Z. J. Public Health* 21:631–637, 1997.
28. MARTINEZ-GONZALEZ, M. A., J. A. MARTINEZ, F. B. HU, M. J. GIBNEY, and J. KEARNEY. Physical inactivity, sedentary lifestyle and obesity in the European Union. *Int. J. Obes.* 23:1192–1201, 1999.
29. MICHELS, T. C. Predicting exercise in older Americans: using the theory of planned behavior. *Mil. Med.* 163:524–529, 1998.
30. MITCHELL, S. A., and R. S. OLDS. Psychological and perceived situational predictors of physical activity: a cross-sectional analysis. *Health Educ. Res.* 14:305–313, 1999.
31. OMAN, R. F., and A. C. KING. Predicting the adoption and maintenance of exercise participation using self-efficacy and previous exercise participation rates. *Am. J. Health Prom.* 12:154–161, 1998.
32. OWEN, N., E. LESLIE, J. SALMON, and M. J. FOTHERINGHAM. Environmental determinants of physical activity and sedentary behavior. *Exerc. Sport Sci. Rev.* 28:153–158, 2000.
33. RODGERS, W. M., and L. GAUVIN. Heterogeneity of incentives for physical activity and self-efficacy in highly active and moderately active women exercisers. *J. Appl. Soc. Psychol.* 28:1016–1029, 1998.
34. ROSS, C. E. Walking, exercising, and smoking: does neighborhood matter? *Soc. Sci. Med.* 51:265–274, 2000.
35. RUCHLIN, H. S., and M. S. LACHS. Prevalence and correlates of exercise among older adults. *J. Appl. Gerontol.* 18:341–357, 1999.
36. SALLIS, J. F., and N. OWEN. *Physical Activity and Behavioral Medicine*. Thousand Oaks, CA: Sage Publications, 1999, pp. 110–134.
37. SALLIS, J. F., and N. OWEN. Ecological models. In: *Health Behavior and Health Education: Theory, Research, and Practice*, 2nd Ed., K. Glanz, F. M. Lewis, and B. K. Rimer (Eds.). San Francisco: Jossey-Bass, 1997, pp. 403–424.
38. SALMON, J., A. BAUMAN, D. CRAWFORD, A. TIMPERIO, and N. OWEN. The association between television viewing and overweight among Australian adults participating in varying levels of leisure time physical activity. *Int. J. Obes.* 24:600–606, 2000.
39. SALMON, J., N. OWEN, A. BAUMAN, M. K. H. SCHMITZ, and M. BOOTH. Leisure-time, occupational, and household physical activity among professional, skilled, and less-skilled workers and homemakers. *Prev. Med.* 30:191–199, 2000.
40. SIMONICK, E., J. M. GURALNIK, and L. P. FRIED. Who walks? Factors associated with walking behavior in disabled older women with and without self-reported walking difficulty. *J. Am. Geriatr. Soc.* 47:672–680, 1999.
41. STERNFELD, B., B. E. AINSWORTH, and C. P. QUESENBERRY, Jr. Physical activity in a diverse population of women. *Prev. Med.* 28:313–323, 1999.
42. SULLUM, J., M. M. CLARK, and T. K. KING. Predictors of exercise relapse in a college population. *J. Am. Coll. Health* 48:175–180, 2000.
43. TAYLOR, W. C., S. N. BLAIR, S. S. CUMMINGS, C. C. WUN, and R. M. MALINA. Childhood and adolescent physical activity patterns and adult physical activity. *Med. Sci. Sports Exerc.* 31:118–123, 1999.
44. UNITED STATES DEPARTMENT OF HEALTH AND HUMAN SERVICES. *Physical activity and health: A report of the Surgeon General*. Atlanta, GA: U.S. Department of Health and Human Services, Centers for Disease Control and Prevention, National Center for Chronic Disease Prevention and Health Promotion, S/N 017-023-00196-5, 1996, pp. 3–8.
45. WILCOX, S., C. CASTRO, A. C. KING, R. HOUSEMANN, and R. C. BROWNSON. Determinants of leisure time physical activity in rural compared with urban older and ethnically diverse women in the United States. *J. Epidemiol. Community Health* 54:667–672, 2000.
46. YIN, Z., and M. P. BOYD. Behavioral and cognitive correlates of exercise self-schemata. *J. Psychol.* 134:269–282, 2000.